Eyal Weiss — *Curriculum vitae*

Contact Information
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Research Interests
Artificial intelligence, automated planning, heuristic search, robotics, control theory

Research Positions

Technion - Israel Institute of Technology, Haifa, Israel

Postdoctoral Scholar, Computer Science, October 2024–current

- Topics: Search, Planning, Motion Planning, Robotics
- Host: Host: Prof. Oren Salzman

Education

Bar-Ilan University, Ramat Gan, Israel

Ph.D., Computer Science, October 2019 to September 2024

- Thesis Topic: *Graph Search and Automated Planning with Multiple Cost Estimators*
- Advisor: Prof. Gal A. Kaminka

Tel Aviv University, Tel Aviv, Israel

M.Sc., Electrical Engineering, March 2018

- Topic: *Minimal Controllability and Minimal Observability of Conjunctive Boolean Networks*
- Advisor: Prof. Michael Margaliot
- *Cum Laude*

B.Sc., Electrical Engineering, October 2016

- *Summa Cum Laude*

Additional Research Experience

Research Assistant — April 2018 to September 2019
Department of Electrical Engineering,
Tel Aviv University
Supervisor: Prof. Michael Margaliot

Honours and Awards

- SoCS 2025 Best Paper Award
- "Adams Fellowship" scholarship for outstanding doctoral students in the Natural and Exact Sciences — 2022–2024
- Excellence scholarship for research students, Computer Science Department — 2023
- "Milgat Hanasi" scholarship for excellent Ph.D. students — 2021–2023
- "Shlomo Shmeltzer Institue for Smart Transportation Scholarship" — 2017
- Excellent student prize for M.Sc. students, Faculty of Engineering — 2017
- "Velger Prize" of excellence for graduate studies in Systems and Control — 2017
- "Final Prize" for excellent students in Electrical Engineering — 2015
- "Freescale Scholarship" for excellent students in Electrical Engineering — 2014
- Dean's Honors (every year of the B.Sc.) — 2012–2016

Hardware and Software Skills

Computer Programming:
- Proficient: C, C++, Java, Python, Matlab, Bash
- Can handle (using AI coding models) any programming language

Laboratory Experience:
- Robotic systems
- Electronics and low-level controllers

Service

Reviewer

- *International Conference on Automated Planning and Scheduling (ICAPS)*
- *Annual AAAI Conference on Artificial Intelligence (AAAI)*
- *International Joint Conference on Artificial Intelligence (IJCAI)*
- *Robotics: Science and Systems (RSS)*
- *IEEE Robotics and Automation Letters (RA-L)*
- *International Conference on Artificial Neural Networks (ICANN)*
- *Automatica*
- *IEEE Transactions on Automatic Control*
- *European Journal of Control*
- *IEEE Transactions on Network Science and Engineering*
- *Optimization and Engineering*

Organizing Committee

- *ICAPS Workshop on Reliability In Planning and Learning (RIPL)*, 2026
- *ICAPS Workshop on Heuristics and Search for Domain-independent Planning (HSDIP)*, 2025
- *ICAPS Workshop on Reliable Data-Driven Planning and Scheduling (RDDPS)*, 2024
- *ICAPS Workshop on Reliable Data-Driven Planning and Scheduling (RDDPS)*, 2023

Session chair

- *European Conference on Artificial Intelligence (ECAI)*, 2023
- *IEEE Conference on Decision and Control (CDC)*, 2018

Journal Publications

1. **E. Weiss** and M. Margaliot, "A Generalization of Linear Positive Systems with Applications to Nonlinear Systems: Invariant Sets and the Poincare-Bendixon Property", *Automatica*, 123, 109358, 2021.

2. **E. Weiss** and M. Margaliot, "Is My System of ODEs k-Cooperative?", *IEEE Control Systems Letters*, 5:73–78, 2020.

3. **E. Weiss** and M. Margaliot, "A Polynomial-Time Algorithm for Solving the Minimal Observability Problem in Conjunctive Boolean Networks", *IEEE Transactions on Automatic Control*, 64(7):2727–2736, 2018.

4. **E. Weiss** and M. Margaliot, "Output Selection and Observer Design for Boolean Control Networks: A Sub-Optimal Polynomial-Complexity Algorithm", *IEEE Control Systems Letters*, 3:210–215, 2018.

5. **E. Weiss** and M. Margaliot and G. Even, "Minimal Controllability of Conjunctive Boolean Networks is NP-Complete", *Automatica*, 92:56–62, 2018.

Conference Publications

1. Y. Wang and **E. Weiss** and B. Mu and O. Salzman, "Bidirectional Search while Ensuring Meet-In-The-Middle via Effective and Efficient-to-Compute Termination Conditions", *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 8987–8995, 2025.

2. Y. Sherma and **E. Weiss** and O. Salzman, "From Agent Centric to Obstacle Centric Planning: A Makespan-Optimal Algorithm for the Multi-Agent Warehouse Rearrangement Problem", *Proceedings of the International Symposium on Combinatorial Search (SoCS)*, 136–144, 2025.

3. D. Gnad and L. Alon and **E. Weiss** and A. Shleyfman, "PDBs Go Numeric: Pattern-Database Heuristics for Simple Numeric Planning", *Proceedings of the AAAI Conference on Artificial Intelligence*, 26507–26515, 2025.

4. **E. Weiss** and A. Felner and G. A. Kaminka, “Tightest Admissible Shortest Path”, *Proceedings of the International Conference on Automated Planning and Scheduling (ICAPS)*, 643–652, 2024.
5. **E. Weiss** and A. Felner and G. A. Kaminka, “A Generalization of the Shortest Path Problem to Graphs with Multiple Edge-Cost Estimates”, *Proceedings of the European Conference on Artificial Intelligence (ECAI)*, 2607–2614, 2023.
6. **E. Weiss** and G. A. Kaminka, “Planning with Multiple Action-Cost Estimates”, *Proceedings of the International Conference on Automated Planning and Scheduling (ICAPS)*, 427–437, 2023.
7. **E. Weiss** and M. Margaliot, “A Generalization of Linear Positive Systems”, *2019 27th Mediterranean Conference on Control and Automation (MED)*, 340–345, 2019.
8. **E. Weiss** and M. Margaliot, “A Generalization of Smillie’s Theorem on Strongly Cooperative Tridiagonal Systems”, *2018 IEEE Conference on Decision and Control (CDC)*, 3080–3085, 2018.
9. **E. Weiss** and M. Margaliot, “Observability Analysis and Observer Design for Boolean Control Networks: A Sub-Optimal Polynomial-Complexity Algorithm”, *IEEE International Conference on the Science of Electrical Engineering*, 2018.

Workshop Publications

1. D. Gnad and L. Alon and **E. Weiss** and A. Shleyfman, “PDBs Go Numeric: Pattern-Database Heuristics for Simple Numeric Planning”, *ICAPS’24 Workshop on Heuristics and Search for Domain-Independent Planning (HSDIP)*, 2024.
2. **E. Weiss** and A. Felner and G. A. Kaminka, “A Generalization of the Shortest Path Problem to Graphs with Multiple Edge-Cost Estimates”, *ICAPS’23 Workshop on Reliable Data-Driven Planning and Scheduling (RDDPS)*, 2023.
3. **E. Weiss** and G. A. Kaminka, “Planning with Dynamically Estimated Action Costs”, *ICAPS’22 Workshop on Reliable Data-Driven Planning and Scheduling (RDDPS)*, 2022.
4. **E. Weiss** and G. A. Kaminka, “Position Paper: Online Modeling for Offline Planning”, *ICAPS’22 Workshop on Reliable Data-Driven Planning and Scheduling (RDDPS)*, 2022.

Teaching Experience and Student Supervision

Positions at the Faculty of Computer Science, Technion – Israel Institute of Technology:

Student Supervisor — 2025–2026
  Supervision of graduation projects of Computer Science students, in the field of heuristic search

Positions at the Department of Computer Science, Bar-Ilan University:

Student Supervisor — October 2022 to March 2024
  Supervision of students conducting research projects at “Alpha” – a research program for gifted high-school students in Israel, associated with the Future Scientists Center

Teaching Assistant — Fall and spring semesters 2024 (current)
  Artificial Intelligence

Teaching Assistant — Spring semester 2023
  Statistical Methods in Computer Science

Teaching Assistant — Spring semesters 2021–2022
  Introduction to Intelligent, Smart and Cognitive Systems

Teaching Assistant — Spring semester 2021
  Object Oriented Programming

Teaching Assistant — Fall semesters 2020–2022
Computer Structure

Positions at Tel Aviv University:

Teaching Assistant — Fall semester 2019–2020
Stochastic Processes
Graduate School of Electrical Engineering

Teaching Assistant — Spring semesters 2017–2020
Signals and Systems
Electrical Engineering International Program

Teaching Assistant — Fall semesters 2017–2020
Random Signals and Noise
Electrical Engineering International Program

Laboratory Instructor — Fall semesters 2016–2020
Laboratory of Advanced Control
Department of Electrical Engineering

Student Supervisor — October 2018 to September 2019
Supervision of graduation projects of Electrical Engineering students, in the field of autonomous driving, using mini-scale cars
Department of Electrical Engineering

Teaching Assistant — Fall semester 2016–2017
Optimal Control
Graduate School of Electrical Engineering

Professional Experience

Positions at Mellanox (today part of Nvidia)

- Network architecture intern — 2015
- Chip design intern — 2013–2014

Social Activities

- Tutor at the economic empowerment program of “Misdar Dorshei Tov” — 2018–2022
- Technical editor at project “Good to Know” of Tel Aviv University — 2017
- Personal mentor at the Children’s Home “Shimony”, at “HaKfar HaYarok”, within the framework of “Youth at risk” — 2012–2014
- Guidance and management roles at the Scouts movement — 2004–2006

Languages

- Hebrew – native
- English – full professional proficiency